Pokoin Whitepaper

Network, Marketplace, and Utility Architecture

Pokoin / PokoinPoS

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Contents

Abstract

Pokoin is a dual-surface ecosystem composed of:

- **PokoinPoS**, a native Proof-of-Stake blockchain with `PKN` as its native coin.

- **CardVault on pokoin.com**, a marketplace and wallet application that uses `PKN` for settlement and user utility.

The project combines public web distribution (Vercel-hosted app and APIs), public chain interfaces (RPC/health/explorer endpoints), and validator/node operations (Oracle/Docker deployment model). This document summarizes architecture, utility, token representations, security posture, and roadmap in one consolidated technical whitepaper.

Vision and Scope

Pokoin focuses on a practical utility stack where blockchain, marketplace, and wallet are tightly integrated:

- card discovery and marketplace operations through `https://pokoin.com/marketplace`;
- wallet and transfer flows through `https://pokoin.com/wallet`;
- chain visibility through `https://pokoin.com/scan` and explorer surfaces;
- public RPC and validator telemetry through `https://rpc.pokoin.com`.

The primary objective is to provide a verifiable, user-facing environment where blockchain settlement and collectible-market workflows coexist in a production-style architecture.

Network Overview (PokoinPoS)

Core Parameters

- Network name: `PokoinPoS`
- Chain ID / Network ID: `26062026` (`0x18dacca`)
- Native asset: `PKN`
- Decimals: `18`
- Public RPC: `https://rpc.pokoin.com/rpc`
- Public explorer host: `https://explorer.pokoin.com`
- Public website: `https://pokoin.com`

Consensus Model

PokoinPoS uses slot-based Proof-of-Stake with signature-backed lottery selection and deterministic validation. Validators with positive balance participate proportionally in block probability, while zero-balance peers can remain connected for network synchronization.

The chain runtime includes:

- slot-based production;
- block and transaction validation;
- longest-chain ordering;
- deterministic ledger reconstruction;
- rollback observation/finality-depth behavior.

System Architecture

Hosting Boundary

Pokoin separates web delivery from chain runtime:

- **Vercel**: website/UI, static metadata, and marketplace API functions on `pokoin.com`.
- **Node infrastructure (Oracle/Docker)**: P2P runtime, validator process, and RPC gateway on `rpc.pokoin.com`.

This boundary keeps user web surfaces elastic and independently deployable while preserving stable chain execution and RPC behavior.

Data Ownership

- **Oracle Postgres**: marketplace catalog/search/event analytics and hot-blueprint rollups.
- **Firebase**: user identity/profile balances/orders/carts and mutable application state.
- **PokoinPoS nodes**: chain state, validator operation, and RPC responses.

Asset Model: PKN and wPKN

Native Coin (PKN)

`PKN` is a native protocol-level coin on PokoinPoS. It is not an ERC-20/BEP-20 contract on a third-party chain and has no token contract address on PokoinPoS itself.

Primary utility includes:

- native transfers and settlement;
- validator reward accounting;
- wallet-compatible chain interactions;
- ecosystem usage in marketplace and chain-facing tools.

Wrapped Representation (wPKN on BNB Chain)

For external DeFi and discoverability, Pokoin maintains `wPKN` on BNB Smart Chain:

- Token: `Wrapped PKN (wPKN)`
- Contract: `0x91A17E2bddfF839078BD395482B38e4AC15276f4`
- Supply model: fixed at `2,000,000 wPKN` for current launch
- Ownership: renounced
- Liquidity surface: PancakeSwap pair

The reserve manifest (`wpkn-reserve.json`) documents reserve discipline and public verification metadata.

Marketplace and Wallet Utility

The web application (`pokoin.com`) provides:

- catalog, search, card detail routes and seller listings;
- wallet route with transfer/swap-oriented UX;
- API-backed homepage snapshots and market hotness scoring;
- scan/explorer route for public chain-facing visibility.

Marketplace APIs include card catalog, versions, search candidates, autocomplete, event capture, and rolling blueprint analytics.

Swap and Exchange Layer

PokoinSwap introduces a native AMM direction centered on `PKN-WPKN` semantics in PokoinPoS runtime, while CardVault includes bridge-style crypto-to-PKN and PKN-to-crypto workflows with quote verification and settlement gating.

At a high level:

- native AMM logic is deterministic and validator-replayed;
- external asset conversions are routed through authenticated backend flows;
- liquidity and payout safety checks gate completion state.

Security and Compliance Baseline

Current documented controls include:

- host hardening scripts (SSH key-only, firewall policy, fail2ban, unattended updates);
- systemd isolation controls (`NoNewPrivileges`, filesystem protections, constrained writable paths);
- runtime secret separation through environment files;
- operational endpoints for health/readiness/status/metrics;
- token-protected operator/admin paths when configured.

Roadmap controls identified by the project include:

- stronger key management and rotation;
- TLS/auth/rate-limit hardening for high-risk interfaces;
- immutable audit trail and formal incident response artifacts;
- CI security gates (dependency scanning, SBOM, fuzz/replay suites).

Market Status and Evidence

Project evidence currently includes:

- live public web surfaces and wallet route;
- live RPC/health/status endpoints;
- public validator/network metadata;
- wrapped token contract and public liquidity pair;
- reserve manifest and metadata artifacts;
- Oracle-backed marketplace analytics pipelines.

The project identifies itself as being in an early market-formation stage, with infrastructure live and reporting maturity increasing over time.

Public Verification Endpoints

- Website: https://pokoin.com
- Marketplace: https://pokoin.com/marketplace
- Wallet: https://pokoin.com/wallet
- Scan UI: https://pokoin.com/scan
- RPC: https://rpc.pokoin.com/rpc

- Node health: https://rpc.pokoin.com/health
- Chain status: https://rpc.pokoin.com/chain/status
- Explorer host: https://explorer.pokoin.com
- Bootstrap manifest: https://pokoin.com/bootstrap-peers.json

Roadmap Direction

Priority roadmap themes:

1. expand measurable marketplace activity and recurring reporting;
2. deepen tokenized-card and liquidity-backed utility paths;
3. harden production controls for broader partner/compliance review;
4. continue validator/network resilience through manifest-driven bootstrap evolution and uptime qualification.

Conclusion

Pokoin presents a utility-focused architecture where a native PoS chain (`PKN`) and a production-style marketplace/wallet web stack are deployed as one ecosystem. The project already exposes public interfaces, operational endpoints, and market-facing token infrastructure, with a documented path toward stronger reporting, security hardening, and broader ecosystem integration.

Source Basis

This whitepaper was compiled from current project documentation, including:

- `README.md` (PokoinPoS repository)
- `docs/public-network.md`
- `docs/website.md`
- `docs/wpkn-bnb-pancakeswap.md`
- `docs/pokoin-swap.md`
- `docs/pkn-native-asset-self-assessment.md`
- `docs/pokoin-market-performance-supporting-documents.md`
- `docs/operations/security-compliance.md`